

NIH Biographical Sketch Common Form

Name: Anderson, Brian Mark

Persistent Identifier (PID) of the Senior/Key Person: <https://orcid.org/0000-0002-2748-1444>

Position Title: Assistant Professor, Medical Physics

Organization and Location: University of California, San Diego, San Diego, California, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
University of California San Diego, San Diego, CA, United States	Resident	07/2021	06/2023	Therapy Physics
The University of Texas MD Anderson UTHealth Graduate School of Biomedical Sciences, Houston, TX, United States	Doctor of Philosophy (PHD)	08/2017	06/2021	Medical Physics
The University of Texas MD Anderson UTHealth Graduate School of Biomedical Sciences, Houston, Texas, United States	Master of Science (MS)	08/2015	07/2017	Medical Physics
Georgia Institute of Technology, Atlanta, GA, US	B.S Nuclear and Radiological Engineering	08/2010	05/2015	Nuclear Engineering

Appointments and Positions

2025 - present Assistant Professor, Medical Physics, University of California, San Diego, San Diego, California, United States

2025 - present Adjunct Assistant Professor, Medical Physics, University of North Carolina, Chapel Hill, Chapel Hill, North Carolina, United States

2023 - 2025 Assistant Professor, Medical Physics, The University of North Carolina, Chapel Hill, Chapel Hill, North Carolina, United States

Products

Products Closely Related to the Proposed Project

1. Anderson BM, Wahid KA, Brock KK. Simple Python Module for Conversions Between DICOM Images and Radiation Therapy Structures, Masks, and Prediction Arrays. Pract Radiat Oncol. 2021 May-Jun;11(3):226-229. PubMed Central PMCID: [PMC8102371](https://pubmed.ncbi.nlm.nih.gov/PMC8102371/).
2. Anderson BM, Rigaud B, Lin YM, Jones AK, Kang HC, Odisio BC, Brock KK. Automated segmentation of colorectal liver metastasis and liver ablation on contrast-enhanced CT images. Front Oncol. 2022;12:886517. PubMed Central PMCID: [PMC9403767](https://pubmed.ncbi.nlm.nih.gov/PMC9403767/).
3. Anderson BM, Lin EY, Cardenas CE, Gress DA, Erwin WD, Odisio BC, Koay EJ, Brock KK. Automated Contouring of Contrast and Noncontrast Computed Tomography Liver Images With Fully Convolutional Networks. Adv Radiat Oncol. 2021 Jan-Feb;6(1):100464. PubMed Central PMCID: [PMC7807136](https://pubmed.ncbi.nlm.nih.gov/PMC7807136/).
4. Anderson BM, Lin YM, Lin EY, Cazoulat G, Gupta S, Kyle Jones A, Odisio BC, Brock KK. A novel use of biomechanical model-based deformable image registration (DIR) for assessing colorectal liver metastases ablation outcomes. Med Phys. 2021 Oct;48(10):6226-6236. PubMed Central PMCID: [PMC9380122](https://pubmed.ncbi.nlm.nih.gov/PMC9380122/).

Other Significant Products Highlighting Contributions to Science

1. Anderson B, Padilla L, Ryckman J, Covington E, Hong D, Woods K, Katz M, Zuhour R, Estes C, Moore K, Bojchko C. Open RT Structures: A Solution for TG-263 Accessibility. International Journal of Radiation Oncology*Biography*Physics. 2024 March; 118(3):859-863. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0360301623079555> DOI: 10.1016/j.ijrobp.2023.09.041
2. Anderson B, Bojchko C. DICOM attribute manipulation tool: Easily change frame of reference, series instance, and SOP

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

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NIH BIOGRAPHICAL SKETCH SUPPLEMENT

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Personal Statement

I am uniquely positioned to contribute to the Big Data Radiation Oncology Collaboration (BigROC) through my combined expertise in radiation oncology data science, multi-institutional AI development, and clinical medical physics practice.

My research trajectory has consistently focused on developing AI solutions for radiation oncology that require large-scale data integration. During my doctoral work at MD Anderson, I developed deep learning methods for liver tumor segmentation and deformable image registration, work that earned the Alfred G. Knudson Jr. Outstanding Dissertation Award.

I have specific expertise in the technical challenges BigROC aims to address. I facilitated automated segmentation algorithms for head and neck cancer MRI and oropharyngeal clinical target volumes using three-dimensional convolutional neural networks. This direct experience with head and neck auto-segmentation, BigROC's demonstrative application, positions me to contribute meaningfully to Aim 3's tumor auto-contouring algorithm development.

My publication record includes over 20 peer-reviewed papers on deep learning applications in radiation oncology, with particular emphasis on segmentation tasks that mirror BigROC's goals.

My technical work on data infrastructure is directly relevant to Aims 1 and 2. Large-scale AI projects live or die by the quality of their data pipelines, and I have dedicated substantial effort to building tools that address precisely the challenges BigROC will face. I developed and published a Python module for DICOM-RT manipulation that converts between DICOM imaging data, structures, dose, and the array formats required for neural network training. This open-source tool streamlines the critical pathway from clinical data to AI-ready datasets. I have built standalone DICOM manipulation applications, operations essential for properly registering multimodal imaging data such as 4D-CT, MRI, and PET across diverse institutional systems.

At UNC, I created DICOM transfer servers enabling seamless data movement between treatment planning systems and clinical infrastructure, solving interoperability challenges that plague multi-vendor environments. These tools have been deployed clinically and represent practical solutions to the data harmonization, and format standardization challenges central to BigROC's mission.

My membership on AAPM Task Group 263, which focuses on standardizing nomenclatures in radiation oncology, provides direct insight into the variability in structure naming conventions across institutions: a key challenge for automated curation. I co-authored work on TG-263 accessibility and surveyed the field on implementation roadblocks, experience that will inform BigROC's approach to nomenclature harmonization.

As a Co-Investigator on an AHRQ R01 examining AI's impact on healthcare safety at UNC, I understand the regulatory, safety, and implementation considerations essential for multi-institutional research. My clinical role as a board-certified medical physicist ensures that BigROC's outputs remain grounded in clinical utility.

My collaborative approach aligns with BigROC's governance model. I have contributed to large team science efforts, including the Radiation Planning Assistant project spanning multiple international institutions. I have presented workshops on AI implementation at AAPM, WIMP, and MD Anderson, demonstrating my commitment to knowledge sharing that will strengthen BigROC's collaborative culture.

Honors

2022	Jack Krohmer Early Career Investigator Competition Winner, American Association of Physicists in Medicine (AAPM)
2022	Alfred G. Knudson Jr. Outstanding Dissertation Award, MD Anderson

2021	Practical Big Data Workshop, Early Career Investigator - Impact Award, American Association of Physicists in Medicine
2019	Science Council Session Selection, American Association of Physicists in Medicine
2018	People's Choice Award for Medical Physics Slam, American Association of Physicists in Medicine
2018	Medical Physics Slam Winner, South West American Association of Physicists in Medicine
2018	Young Investigator Award, Southwest American Association of Physicists in Medicine
2018	Early Career Medical Physicist Award, Winter Institute of Medical Physics

Contributions to Science

1. Open-source tools for DICOM data conversion and deep learning in radiation oncology

A major barrier to the adoption of deep learning in radiation oncology is the complexity of converting between clinical DICOM data formats and the array-based representations required by AI frameworks. To address this, I developed DicomRTTool, an open-source Python module that provides bidirectional conversion between DICOM images, RT structures, NumPy arrays, and SimpleITK images (Anderson, et al., Pract Radiat Oncol, 2021). At the time of publication, no purpose-built Python module existed for converting deep learning prediction arrays back into DICOM RT structures, a critical step for deploying segmentation models in clinical workflows. DicomRTTool also provides intuitive methods for rapid data curation of RT-structure files, including identification of unique region-of-interest names across heterogeneous datasets, flexible name aliasing, and batch processing capabilities.

The tool is designed to serve researchers and clinicians at all experience levels, from those new to deep learning who need straightforward data preparation pipelines to experienced developers building auto-segmentation and radiomic analysis workflows. Since its release, DicomRTTool has been widely adopted across the radiation oncology research community, accumulating over 100 GitHub stars, 30 repository forks, and over 1,000 downloads via PyPI.

The tool has been cited in studies spanning auto-segmentation of multiple disease sites, adaptive radiotherapy, synthetic image generation, and radiomic feature extraction. By lowering the technical barrier to entry for AI research in radiation oncology, DicomRTTool has enabled researchers at institutions worldwide to focus on model development rather than data preprocessing, accelerating the pace of clinically relevant AI research.

2. Enabling standardized nomenclature adoption in radiation oncology clinics

Standardized nomenclature is foundational to data sharing, workflow automation, and outcomes research in radiation oncology, yet widespread adoption of the AAPM TG-263 standard has been hindered by the considerable burden of creating and managing compliant structure template libraries.

A national survey by TG-263U1, on which I served as a member, found that while 83% of respondents considered standardized nomenclature important, staffing and implementation effort were the primary barriers delaying adoption, particularly at smaller and community clinics (Covington et al., J Appl Clin Med Phys, 2024).

To directly address this gap, I led the development of Open RT Structures, a freely available, open-source program that enables clinics to generate TG-263 compliant structure templates compatible with multiple treatment planning systems (Anderson et al., Int J Radiat Oncol Biol Phys, 2024).

The tool was designed to serve the full spectrum of radiation oncology professionals, including physicists, dosimetrists, and physicians, and was built using the .NET framework to ensure broad accessibility without requiring programming expertise. Open RT Structures eliminates the need for each clinic to independently build and maintain template libraries, which had been identified as a significant resource burden especially for community practices and low-resource settings.

By providing a vendor-neutral solution that generates templates in multiple formats, the tool facilitates consistent contour

naming across institutions, which is a prerequisite for multi-institutional big data efforts, AI model training, and meaningful toxicity and outcomes research. This work directly supports the infrastructure needed for large-scale data curation and sharing initiatives in radiation oncology.

3. Simple and Safe: DICOM attribute modification for image registration in radiotherapy

In radiation oncology, the integration and registration of multiple imaging modalities is essential for accurate diagnosis and treatment planning. Images acquired within the same study, such as multiple MRI sequences in stereotactic radiosurgery, share the same Frame of Reference UID, which prevents correction of inter-scan motion through re-registration. To address this, DICOM attributes must be modified, but doing so typically requires expert knowledge of DICOM file structure or specialized commercial software that may inadvertently alter other attributes, potentially introducing clinical errors.

I developed and published a standalone DICOM Attribute Manipulation Tool that allows users to safely and selectively modify the three most commonly edited DICOM attributes: Frame of Reference UID, Series Instance UID, and SOP Instance UID (Anderson, Bojchko, J Appl Clin Med Phys, 2025). The tool features a graphical user interface requiring no programming expertise, making it accessible to physicists, dosimetrists, and therapists across clinical settings.

It is particularly valuable for 4D-CT workflows, where the Frame of Reference must be changed consistently across all phases of the scan, a task that is error-prone when performed manually. The tool has been deployed clinically at multiple institutions, replacing ad-hoc and error-prone manual processes. It is freely available as open-source software, lowering barriers for clinics of all sizes, including community and low-resource practices.

By simplifying a routine but technically challenging task, this tool reduces the likelihood of registration errors that contribute to treatment uncertainty, ultimately improving the precision of image-guided radiation therapy.

4. Deep learning segmentation and AI for clinical workflows across multiple disease sites

Throughout my career, I have developed and applied deep learning segmentation and AI methods across multiple disease sites to address unmet clinical needs in radiation oncology and interventional radiology, spanning algorithm development through clinical deployment.

My doctoral work at MD Anderson focused on improving local liver ablation therapy through deep learning and biomechanical modeling. I developed automated segmentation of liver, colorectal liver metastases, and ablation zones on both contrast and non-contrast CT (Anderson et al., Adv Radiat Oncol, 2020; Anderson et al., Front Radiat Oncol, 2022). These tools enabled near real-time assessment of ablation coverage during interventional procedures and directly contributed to a Phase 2 clinical trial (NCT04083378). This work received the MD Anderson Knudson Outstanding Dissertation Award.

I contributed to deep learning auto-segmentation for cervical cancer adaptive radiotherapy, where our models enabled online dose optimization through rapid organ-at-risk segmentation (Rigaud, Anderson et al., IJROBP, 2020), and to auto-delineation of oropharyngeal clinical target volumes using 3D CNNs (Cardenas, Anderson et al., Phys Med Biol, 2018), an early demonstration of deep learning for target volume prediction. More recently at UNC, I have applied deep learning to predict dysphagia outcomes and elective nodal volumes in head and neck radiotherapy, tools designed to support peer review and reduce treatment planning variability.

I also developed a CNN for predicting in-vivo EPID transit images from patient anatomy and plan data (Anderson et al., Med Phys, 2024), providing a novel approach to on-treatment error detection. This body of work demonstrates a consistent record of applying AI to solve clinically meaningful problems across disease sites and treatment modalities.

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